

Software Requirements and Design Document

For

Group 13

Version 2.0

Authors:

Jasper, Owen, Hubert

Overview

This project is for the development of a full-stack mobile and web application designed to increase public awareness and engagement with all aspects of politics. Politics often determines important aspects of many people's lives; however, most people do not have the time or the need to go through pages of bills and actions that are taken by their local and state officials.

Our application will serve as a personalized civic hub, allowing users to view their representatives, track legislation, simulate voting on bills, and evaluate representative accountability. By using an intuitive design and helpful guides throughout the process, we expect many people to begin using the app and have a more direct impact on their local officials. Overall, this app will help people become more involved in politics by making it easy for them to do so.

This document is relatively the same, as we made sure we had a solid framework from the start that we could build upon. No use cases or classes were added or changed.

Functional Requirements

1. The system will provide a Representative Directory with profiles of local, state, and district-level representatives, including voting history and issue stances. (High priority)
2. The system will provide policy summaries and voting simulations, including AI-generated summaries of upcoming bills, and allow users to cast simulated votes. (High priority)
3. The system will calculate and display a Representative Consistency Score, comparing stated policy positions with actual voting records. (High priority)
4. The system will offer a Belief Alignment Quiz to match users with representatives/candidates based on shared policy views. (Medium priority)
5. The system will provide Candidate Transparency Tools, including verified candidate accounts, profiles, stances, and activity feeds. (Medium priority)
6. The system will provide a Personalized Civic Hub that tracks users' votes, quiz results, and aligned representatives, and curates news from verified outlets. (High priority)
7. The system will allow registered users to login to their account through a secure login page, and new users to register on a register page
8. The system will allow Representatives and Candidates to update their profiles and share updates during election cycles. (Medium priority)

Non-functional Requirements

1. **Data Reliability:** The system will ensure accurate and up-to-date legislative and representative information through reliable API integrations.
2. **Bias Mitigation:** The system will ensure that AI-generated summaries remain neutral and factual, requiring monitoring and fine-tuning.
3. **Scalability:** The system will be able to handle real-time legislative data updates across city, county, and state levels.
4. **Performance:** The system will respond to user requests (e.g., quizzes, voting simulations) within 2 seconds under normal loads.
5. **Security:** The system will use secure authentication and data storage for user accounts.
6. **Usability:** The system will feature intuitive mobile-first UX/UI design to encourage user adoption.
7. **Reliability:** The system will have 99% uptime availability for legislative and representative data access.

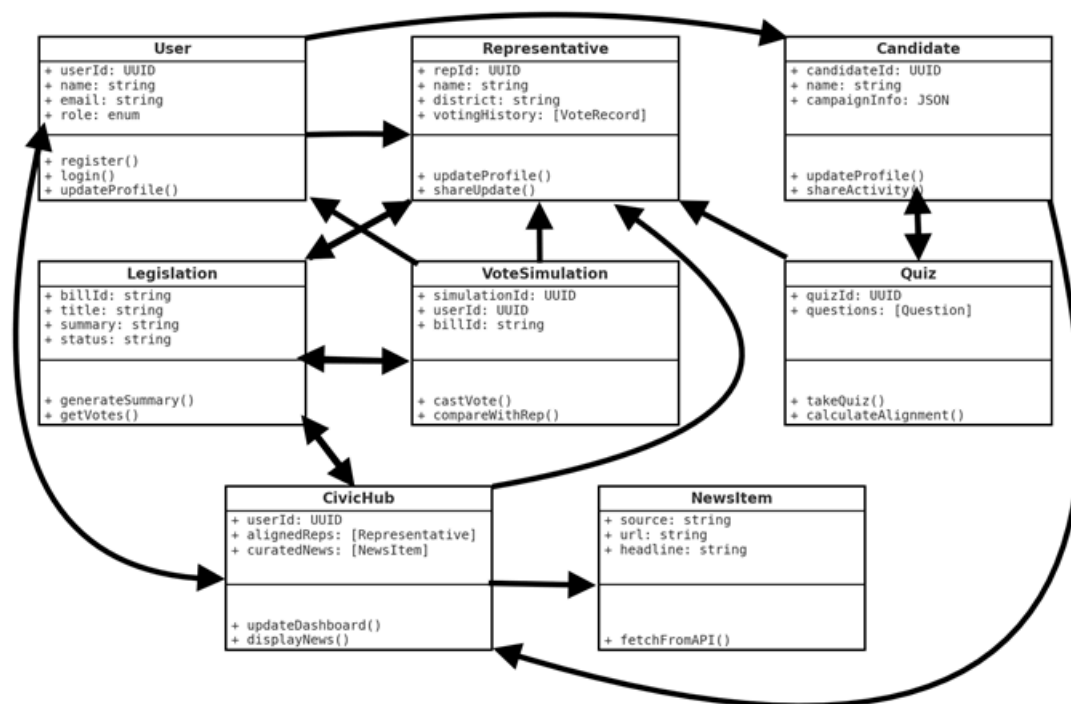
Use Case Diagram

Use Cases:

- View Representative Directory
- View Policy Summaries
- Simulate Voting on Bills
- Calculate Representative Consistency Score
- Take Belief Alignment Quiz
- Manage Candidate/Representative Profiles
- Access Personalized Civic
- Create Account
- Login into Account

Class Diagram and/or Sequence Diagrams

- **User** (attributes: ID, profile, ,password, preferences; methods: login, updateProfile)
- **Representative** (attributes: name, district, stances, voting history; methods: updateProfile, shareUpdate)
- **Candidate** (attributes: name, stances, campaign info; methods: updateProfile, shareActivity)
- **Legislation** (attributes: billID, title, summary, status; methods: generateSummary, getVotes)
- **VoteSimulation** (attributes: userID, billID, simulatedVote; methods: castVote, compareWithRep)
- **Quiz** (attributes: quizID, questions, results; methods: takeQuiz, calculateAlignment)
- **CivicHub** (attributes: userDashboard, curatedNews; methods: updateDashboard, displayNews)



Operating Environment

1. Platforms: Web and Mobile
2. Back-End: C# and .NET
3. Database: MySQL

4. Third-Party APIs: Google Civic Information API, OpenStates API, BallotReady APIs, News APIs, OpenAI/Hugging Face Transformers
5. Operating Systems: Cross-platform (iOS, Android, Windows, macOS, Linux browsers)

Assumptions and Dependencies

· Assumptions:

- APIs such as Google Civic, OpenStates, and News APIs remain accessible and reliable.
- AI models such as OpenAI or Hugging Face will provide neutral, factual summaries when fine-tuned.
- Users will use the app if the UI is intuitive and designed with a mobile-first approach.

· Dependencies:

- Legislative and election data availability from third-party APIs.
- Hosting environment that supports .NET web services and MySQL databases.
- GitHub for version control and collaborative development.